# DEPARTMENT OF CSE (AI & ML)

**Session: June – July 2026**

## PI Labs Summer Internship
## Interim Presentation

# From Conflict to Consensus: Boosting Medical Reasoning via Multi-Round Agentic RAG


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# Problem Statement

Clearly Define the Problem:

- Large Language Models generate logical-sounding but factually incorrect medical advice due to hallucinations.
- Standard architectures lack peer-review or clinical debate mechanisms to verify outputs before reaching patients.
- Vanilla RAG cannot guarantee safety or resolve contradictions when different retrieved journals disagree.

Real-World Relevance:

- Incorrect AI clinical recommendations can lead to life-threatening errors or wrong diagnoses.
- Safety-first guardrails are essential to prevent prescribing incorrect medications or dosages.
- Automatically evaluates consensus and blocks answers when clinical literature is contradictory.
- Saves healthcare professionals time spent manually cross-referencing conflicting medical sources.
- Provides reliable, debate-proven consensus answers for clinical decision support systems.

Scope and Constraints:

# Literature Review / Background

Multi-Agent Debate Frameworks: Prior research shows that setting up LLMs to critique each other in a loop improves logical reasoning, but these models still hallucinate clinical facts when not grounded in external evidence.

Medical RAG Systems (e.g., MedRAG): Existing medical QA systems retrieve clinical literature (from the MedCorp corpus) using static queries, but they lack interactive debate and cannot resolve contradictions when source texts disagree.

Knowledge Gaps Identified: Lack of a unified multi-round framework that dynamically analyzes agent disagreements mid-debate to generate targeted search queries to actively retrieve verifying clinical evidence.

# Methodology / Approach

Zero-Shot Medical Routing: Evaluates user queries to initiate isolated medical debate sessions.

Multi-Round Persona Debate: Medical Expert A (Primary Care), Expert B (Specialist), and Expert C (Acute Care) generate, critique, and refine answers in parallel rounds.

Active Disagreement Retrieval: A Coordinator agent identifies points of dispute, generates precise search queries, queries MedCorp, and reranks results via MedCPT cross-encoder.

Safety Guardian Layer: Evaluates final consensus, scores clinical confidence (High/Medium/Low), and enforces a safety block if confidence is low.

# Work Completed So Far

Retriever Microservice Setup: Deployed local retrieval services running BM25 keyword matching and MedCPT Cross-Encoder reranking.

Medical Agent Pipeline Development: Programmed the 3-expert debate logic, peer critique templates, and moderator synthesis protocols.

Safety Guardian & Consensus Judge: Deployed validation layers to detect factual alignment and audit clinical confidence.

Interactive Testing Interface: Created command-line shells to run and debug multi-round medical debates in real time.

# Preliminary Results / Observations

Collaborative Self-Correction: Witnessed Expert B (Specialist) catch and correct an incorrect drug dosage recommendation suggested by Expert A during Round 2 critique.

Guardian Safety Blocking: The Safety Guardian successfully blocked responses to highly ambiguous or dangerous test cases, returning the exact reason for low confidence.

Rate Limit Adjustments: Configured error handling to manage Groq API rate limits (429 errors) during simultaneous agent invocations.

Evaluation Logs: Verified successful test runs on sample MedMCQA multiple-choice questions.

# Plan for the Remaining Weeks

Batch Evaluation on Dataset: Run the full MedMCQA dataset on the evaluator script (ma_rag_evaluator.py) using Llama-3.1-8B-Instant and Qwen-3-8B.

Analyze Test-Time Scaling: Track and chart how answer accuracy improves as the number of debate rounds and active retrieval loops increases.

Prompt Engineering & Parameter Tuning: Adjust system temperatures and top_k retrieval parameters to optimize answer consensus.

Final Report & Documentation: Compile experimental results and figures, write the final thesis report, and prepare the final presentation.